

Falsification probes separate shared response agreement from perturbation-specific retrieval

A Response agreement

Audit-delta Pearson

K562

RPE1

Mean-effect

Label-shuffled

GEARS

0.0 0.2 0.4 0.6

Audit-delta Pearson

0.0 0.2 0.4 0.6

Audit-delta Pearson

- Mean-effect: target-blind response probe/baseline
- △ Label-shuffled: target labels scrambled diagnostic probe

B Perturbation-specific retrieval

Mean reciprocal rank (MRR)

K562

RPE1

0.00 0.02 0.04 0.06

MRR

0.00 0.02 0.04 0.06

MRR

- GEARS: target-aware predictive model
- | Random-rank reference: HN/N

